Zhewei Gui

Robot Learning · Dexterous Manipulation · Reinforcement Learning · Sim-to-Real

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dex-hot.github.io

Open to full-time roles & PhD opportunities

EDUCATION

National University of Singapore · College of Design and Engineering — Singapore | Aug 2024–Feb 2026
M.Sc. in Robotics

Tongji University · School of Mechanical Engineering — Shanghai | Sep 2020–Jul 2024
B.Eng. in Mechanical Design, Manufacturing and Automation · Multiple merit and community scholarships

RESEARCH

DexHOT: Learning Hand–Object Configuration Transfer for Dexterous Manipulation — Sep 2025–Present
Co-first Author · Prof. Hao Dong's group, Peking University / PrimeBot

- Introduced **hand–object configuration transfer**: a relation-conditioned policy performs closed-loop transfer between configurations jointly specifying hand articulation and object pose relative to the wrist, while hierarchical subgoals compose long-horizon behavior.
- Built DexHOTBench with **120 objects and 1.2M feasible configurations**; one policy achieved **80.5% / 69.4%** transfer success on seen / unseen objects in simulation.
- Completed real-world Sim-to-Real deployment: **61.5%** success over 200 transfers and **90%** success across six demonstration-guided long-horizon tasks, including pouring, scanning, and screw driving.

DexSinGrasp: Learning a Unified Policy for Dexterous Object Singulation and Grasping in Densely Cluttered Environments (RA-L) — Jan–Jun 2025
Second Author · Prof. Shao Lin's group, National University of Singapore

- Built and trained RL policies in Isaac Gym, designed rewards, distilled a vision-based policy with a teacher–student setup, and deployed on LEAP Hand + xArm for robust singulation and grasping in dense clutter.

INDUSTRY EXPERIENCE

JD.com · General Algorithms Group | Research Intern — Beijing | Sep 2025–Jan 2026

- Optimized a dexterous grasping Sim-to-Real framework combining RL and imitation learning, reaching ~**90%** general closed-loop grasp success; supported dynamic targets, target-specific grasping, and human/cross-embodiment retargeting.

AgiBot · Embodied Manipulation Group | Reinforcement Learning Intern — Shanghai | Apr–Sep 2025

- Integrated PPO + DP3 into a generalized grasping Sim-to-Real pipeline; identified arm/hand parameters, optimized latency/performance trade-offs, and migrated the RL stack from Isaac Gym to Isaac Lab.

Fourier Intelligence | Algorithm Intern — Shanghai | Dec 2024–Jan 2025

- Trained stable biped locomotion with HIMPPO in Isaac Gym, tracked commanded velocities, and validated robustness via Isaac Gym → MuJoCo sim-to-sim transfer.

Orient Securities | Quantitative Research Intern — Shanghai | Mar–Aug 2023

- Optimized XGBoost hyperparameters and feature combinations using Hyperopt, achieving a **0.104 RankIC**.

SKILLS & ADDITIONAL

Programming & tools: Python, PyTorch, MATLAB, SQL, Gym, Isaac Gym / Sim / Lab, MuJoCo

Research: Reinforcement learning, imitation learning, dexterous manipulation, Sim-to-Real, cross-embodiment retargeting

Languages: English (TOEFL 103, GRE 329, CET-6); Mandarin Chinese (native)

Interests: Amateur 5-dan Go; former university table tennis and swimming team member